Which Way Does Fairness Go? The Selection Rate Predicts Whether an Attribute-Blind Constraint Gives or Takes

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***Abstract*—A group-fairness constraint is a statement about a gap, and says nothing about which of two very different worlds should close it: one where the disadvantaged group is lifted, and one where the advantaged group is pulled down. Both satisfy the constraint and both score identically on the metric that certifies them. We ask which of the two occurs, and show it is predictable in advance from a quantity every deploying organisation already has — the rate at which its current model says yes. Across 18 independent populations spanning two household surveys, US mortgage records, criminal-justice risk data, US law-school outcomes and the Dutch national census, a parity constraint withdraws favourable decisions below a crossover and extends them above it. On UCI Adult, six in-processing methods each shrink the pool by 7.9–22.1%; on mortgage lending the identical constraint grows it. The certifying metric reports the same success in both cases. The relationship is not an artifact of how the selection rate is varied: holding the task, features, groups and fitted scores fixed and moving only the decision threshold reproduces both the direction and the location of the transition. It survives a twenty-five-fold range of constraint tolerance, a boosted-tree learner, and a protected attribute swapped from sex to race. Crucially, it is bounded: under *post-processing*, which reads the protected attribute at decision time, the effect vanishes entirely ($r = -0.024$ against $+0.585$), and contemporaneous theory says why — in that regime the direction is determined, not distributional. We confirm the theory's prediction for it in 18 of 18 populations. The practical result is a procedure: locate your own crossover by sweeping your deployed model's threshold, needing no new data.**

***Index Terms*—algorithmic fairness, levelling down, demographic parity, in-processing, selection rate, disparate impact**

I. Introduction

A fairness constraint says two groups' rates should be close. It says nothing about how. The gap can close because the disadvantaged group is lifted, or because the advantaged group is pulled down. Both satisfy the constraint; only one is what anybody meant.

That this ambiguity exists is established. Mittelstadt *et al.* [2] made levelling down the centre of a well-known critique; Maheshwari *et al.* [4] report that it "often goes unnoticed in the overall performance of the model"; Ferry *et al.* [3] built an auditing framework because an aggregate score reports neither how many people were affected nor in which direction. We take all of that as given.

Our question is the one this literature leaves open: **when?** A deploying team cannot act on "this sometimes happens". It can act on a rule that says which way its own system will move, before it moves.

A. Scope, stated first

This is a paper about **attribute-blind in-processing** constraints — the regime in which the protected attribute is unavailable at decision time. That is where almost every deployed system sits, because per-group thresholds constitute disparate treatment on their face in most jurisdictions. Under post-processing, which does read the attribute, the effect we report disappears (Section VI). This is not a weakness of the finding but the boundary theory draws, and we verify it from both sides.

B. Contributions

1) **An empirical characterisation of a conditionality that theory predicts.** Contemporaneous theory [5] establishes that in the attribute-blind regime the direction is distribution-dependent. It contains no experiments and its conditions are ordering relations on score regions rather than measurable quantities. We supply the measurement across 18 populations and five domains, and find its stated conditions hold on **0 of 26** arms.
2) **Identification of the operative variable by a design that separates it from its confound.** Varying the selection rate by moving a label threshold cannot distinguish "the rate sets the direction" from "task difficulty does". Holding the task exactly fixed and moving only the decision rule separates them.
3) **A procedure, not an observation.** The same design lets a practitioner locate their own crossover on a deployed model. Four independent estimates of the mortgage crossover agree.

4) **Boundaries stated rather than discovered by a referee**, including a computable test for when the procedure cannot be used at all.

II. Related Work

Whether levelling down is universal. A 2026 theory paper [5] proves in a population-level Bayes framework that the answer depends on the deployment regime. Where the protected attribute is available at decision time, fairness "necessarily (weakly) improves outcomes for the disadvantaged group and (weakly) worsens outcomes for the advantaged group". Where it is not — the attribute-blind regime, which is ours — "the impact of fairness is distribution-dependent: fairness can benefit or harm either group . . . leading to either leveling up or leveling down." **We therefore do not claim the conditionality itself.**

Levelling down, and the remedy. Mittelstadt *et al.* [2] argue that fairness interventions frequently equalise by degrading the better-off group, and *also propose the remedy*: minimum rate constraints requiring "that every group has, at least, a minimal selection rate, precision, or recall". We claim neither the observation nor the remedy as novel. Our contribution relative to that work is narrow and checkable: a scope condition they do not have; in-processing rather than post-processing, so no model reads the attribute when predicting; a population-level floor stacked *with* parity rather than a per-group floor replacing it; and held-out replication where they report training-set results.

Auditing individual impact. Ferry *et al.* [3] propose a framework whose first three dimensions are impact size, change direction and decision rates. That is the decomposition we use throughout, and we claim no part of it. What we add is what those axes report at scale: the change-direction dimension is not a property of the method being audited but of the population it is applied to.

The selection-rate axis. The nearest prior work is Goethals *et al.* [8], which studies the cost of fairness as a function of available budget and reports that "the level of available resources significantly influences this cost, a factor overlooked in previous evaluations". The difference is structural: in their setting positive decisions are a fixed resource, so the total is constant by construction and the directional effect we measure cannot arise.

Fairness gerrymandering and dataset monoculture. Kearns *et al.* [6] show constraints on marginal groups can be satisfied while subgroups are badly treated. Ding *et al.* [7] argue the field should stop drawing conclusions from UCI Adult; we replicate across ACS-derived populations and then find that is *not sufficient*, because they share one survey instrument.

III. Setup and Method

Data. Eighteen independent populations across four instruments: UCI Adult (45,222 rows); ACS Income [7] across twelve US states and two protected attributes; HMDA mortgage records for two states, by race and sex and split by loan purpose; COMPAS [9] (5,278 rows on the race arm after ProPublica's documented filter); LSAC bar passage (20,798 rows); and the Dutch national census of 2001 (60,420 rows).

The last three were added to leave the instruments the finding was formed in. The Dutch census carries a between-group gap of 0.298, roughly twice Adult's, making it the test of the standard alternative explanation. LSAC was chosen because bar passage is naturally generous (base rate 0.890).

Convention. $y = 1$ is always the favourable outcome. This required inverting COMPAS's recorded target, which encodes recidivism, and declaring *Female* the advantaged group on its sex arm, since women in that cohort reoffend less. Both are enforced by automated checks, because either error produces every directional result sign-inverted without raising.

Mitigation. `ExponentiatedGradient` [1] at $\varepsilon = 0.01$ unless stated. No model reads the protected attribute at prediction time except where post-processing is explicitly under test.

Exclusion rules. An arm is discarded if the baseline demographic-parity gap is below 0.05, or if the baseline classifier's accuracy falls below $\max(p, 1-p)$ — the accuracy of always predicting the more common label. The second rule matters more than it sounds (Section IX).

IV. Parity Is Bought by Withdrawal — Where It Is

On UCI Adult, every method in a six-method in-processing ablation reduces the total number of favourable decisions, by 7.9% to 22.1%. Not one closes the gap primarily by extending decisions to the disadvantaged group. Eighteen of nineteen survey arms do the same.

And then it reverses. On HMDA mortgage data the identical constraint *grows* the pool by 4.3%, at an exchange rate of 0.50 — two favourable decisions created for every one destroyed. The parity metric reports 0.018 on the population that destroyed a fifth of its favourable decisions, and 0.010 on the one that created more than it destroyed.

V. What Sets the Direction

A. A single-factor design

ACS Income's label is "earns more than \$50,000" — a benchmark convention, not a fact. Moving it on one fixed state varies the selection rate while holding population, instrument, feature set, group ratio and proxy structure fixed. The change in the pool rises with the baseline selection rate at $r = +0.801$, and $+0.980$ once the confounded between-group gap is partialled out. The sign flips: 22 decisions destroyed per one created at a rate of 0.03, against 0.75 at 0.89.

B. The transition appears without being manufactured

Pooling two states across five real loan products — nothing manipulated — home improvement at a selection rate of 0.555 levels *down* (-1.57%) while refinancing at 0.871 levels *up* ($+2.95\%$), with $r = +0.803$ and Spearman $\rho = +0.900$. A bank running one constraint across its loan book would take opportunities away in one product and hand them out in another, with its fairness report showing success in both.

TABLE I
THE RELATIONSHIP ACROSS INSTRUMENTS

Instrument	Domain	r
ACS Income	income prediction	$+0.801$
HMDA	mortgage lending	$+0.858$
COMPAS	criminal justice	$+0.870$
Dutch census	occupational status	$+0.915$
LSAC	legal education	unsweepable

TABLE II
WHAT EACH MANIPULATION PRESERVES

Manipulation	Direction	Magnitude
Route to the rate	unchanged	differs 3×
Tolerance, 25× range	unchanged	differs 4×
Boosted trees (5/5 pops.)	unchanged	differs 3×
Attribute, sex → race	unchanged	—
Criterion → equalized odds	weaker, unevenly	differs 8×
Optimiser → post-proc.	**vanishes**	—

C. *It is the rate, not the difficulty of the task*

The design above moves the *label*. "Earns over $100,000" is not a rarer version of "earns over $20,000"; it is a harder problem. That design therefore cannot separate two explanations which predict everything observed so far.

We separate them by holding the task completely fixed — same rows, features, groups and *fitted scores* — and moving only where the model draws its line. The relationship reproduces, and so does the crossover's location: on Oregon the two routes give 0.362–0.653 and 0.353–0.637. **Task difficulty is not doing the work.**

With twelve points chosen from each population's viable band, the relationship holds on the Dutch census ($+0.946$), South Carolina ($+0.905$) and COMPAS ($+0.844$). Alabama and Kentucky return *negative* values on eleven and ten arms; their viable bands top out at 0.566 and 0.561, at or below every crossover measured, so both sweeps lie almost entirely on one side of the transition. **We therefore claim the relationship across the crossover and withdraw any claim that it holds within the low-rate region alone.**

D. *Five domains, and the group-gap alternative*

Table I gives the picture. The pre-registered alternative — "this is a property of income prediction and American mortgage lending", requiring $|r| < 0.30$ — loses on both new instruments.

The oldest competing account is that the *gap between groups* drives the direction. The Dutch census has a gap of 0.298, roughly twice Adult's, making it the population where that explanation should win. It gives $r = +0.915$ across all six arms, none excluded.

E. *Robustness*

Table II summarises. **The selection rate predicts which way. It orders how much within a population and does not transfer it between them**: the signed distance from a population's own crossover gives ρ of $+0.96$, $+0.95$ and $+0.78$ in three of four populations, but a pooled slope reaches only $r = +0.487$ against a pre-registered $+0.70$ bar.

TABLE III
CROSSOVER LOCATION, TWELVE-POINT SWEEPS

Population	Domain	Crossover
COMPAS	criminal justice	0.511
South Carolina	income	0.530
Oregon	income	0.558
Dutch census	occupational status	0.576
HMDA (one market)	lending	0.659–0.758

VI. THE REGIME BOUNDARY

The last row of Table II is not fragility. Post-processing reads the protected attribute at prediction time; the reduction does not. That is the distinction on which the theory of [5] is built.

Running post-processing at each population's natural operating point across 17 populations gives $r = -0.024$, against $+0.585$ for the reduction on the same populations. The pre-registered prediction failed and its naive alternative won outright.

In the attribute-aware regime the theory says there is nothing to predict, because the direction is determined. Its prediction for that regime is that the advantaged group loses and the disadvantaged gains, always. On our populations that holds in **18 of 18**, without exception. (This check was decided after the failure and is post-hoc.)

Section IX notes that the same theory's *conditions* hold on 0 of 26 arms. Its regime distinction holds on 18 of 18. The theory is right about which regime you are in mattering, and wrong about which quantity decides it within one — which is what an empirical study can settle and a population-level argument cannot.

VII. BOUNDARIES

A. *The crossover is nearly stable*

Table III shows four populations across three domains, two countries and four instruments agreeing within 0.08, with standard deviation 0.029 and mean 0.544. An earlier version of this work called the crossover "population-specific" from two coarse six-point brackets whose difference was largely imprecision. The instruction is **expect about 0.54 and check**.

Two population properties appeared to explain the residual, at $r = +0.724$ and $+0.865$. They are collinear with each other at $+0.947$ and neither is distinguishable from chance at four populations ($p = 0.277$, $p = 0.135$). We report them as a hypothesis for a larger study, not a formula.

B. *The procedure is unusable on very generous tasks*

Varying the selection rate by moving a threshold works by *degrading* the classifier. On a task where most people already qualify, reaching a low selection rate requires a model that refuses qualified people, and past a point that model is worse than a constant. Defining the *viable band* as the range of selection rates reachable by a classifier still beating $\max(p, 1-p)$: Dutch spans 0.895, COMPAS 0.859, typical ACS ≈ 0.51, and LSAC only **0.056**. On LSAC, where 89% pass the bar, no choice of thresholds produces a usable sweep. This is computable before running anything.

When the viable band forbids a sweep, the crossover can still be located by comparing natural sub-populations — which is how the mortgage crossover was first found.

C. Small populations

Below roughly 2,500 test subjects the method's own randomness exceeds the entire effect of the constraint. COMPAS has 1,584 and flips sign between seeds on two of four arms; LSAC has 6,240 and flips on none. COMPAS carries direction here and never magnitude.

VIII. What to Do About It

Algorithm 1 states the procedure. Three of its steps are worth drawing out, because each is a place where a plausible-looking run returns a meaningless answer.

It can refuse. Lines 6–8 compute the viable band and exit before any constraint is fitted, if no threshold reaches a low selection rate with a model still beating the trivial predictor. On LSAC that band spans 0.056 and the procedure declines, rather than returning arms that would be discarded later.

It can return VOID. Line 16 requires the pool to have moved at all. A correlation computed over arms that barely move is fitted to noise: Connecticut returns $r = -0.924$ on a spread of a tenth of a percentage point, and without this test that number reads as a refutation of the whole finding.

It returns a sign, never a size. The distance from the crossover *orders* the effect within a population, so a team can rank its own segments; but no slope transfers between populations, so lines 25 and 27 return a direction and stop there.

The floor is **not a new method**: a selection-rate floor is a linear moment constraint inside the framework of [1], and a variant of the minimum rate constraints of [2]. It costs about 0.12 accuracy points and moves the exchange rate from 1.47 favourable decisions destroyed per one created to 0.88 across nineteen arms, landing below 1.0 in sixteen of them, with the benefit scaling with the damage at $r \approx -0.99$.

IX. Method, and What We Got Wrong

Every prediction was registered with its thresholds *and the naive alternative it had to beat*, before the data existed. Three episodes matter for reading the results above.

A pre-registered test that failed. The rule does not survive equalized odds at the bar we set: $r = +0.644$ on two states, $+0.334$ on five, against a $+0.70$ threshold. Alabama alone would have given $+0.822$ with every prediction holding.

Two headline results withdrawn. The operating-point design varies the selection rate by degrading the classifier, and nothing bounded how far. Excluding arms beaten by the trivial predictor leaves Alabama with two arms and LSAC with one:

Algorithm 1 Locate the crossover; decide whether to add a floor

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Require: deployed model h, held-out (X, y, a), fairness constraint C
Ensure: predicted direction, and whether a selection-rate floor is needed

1: **if** C does not constrain selection rates **then**
2:     **return** OUT OF SCOPE {e.g. equalized odds}
3: **end if**
4: $\pi \leftarrow \max(\bar{y},\, 1-\bar{y})$ {trivial-predictor accuracy}
5: $s \leftarrow h(X)$
6: $\mathcal{V} \leftarrow \{\tau : \mathrm{acc}(s > \tau) \geq \pi \ \wedge\ 0.05 \leq \overline{(s > \tau)} \leq 0.95\}$
7: **if** $\mathrm{span}(\mathcal{V}) < 0.40$ **then**
8:     **return** UNSWEEPABLE; compare natural sub-populations
9: **end if**
10: $T \leftarrow 12$ thresholds from $\mathcal{V}$, spread evenly in *selection rate*
11: **for all** $\tau \in T$ **do**
12:     $r_\tau \leftarrow \overline{(s > \tau)}$; fit C on the arm thresholded at τ
13:     $\Delta_\tau \leftarrow$ % change in favourable decisions
14:     discard τ if parity gap < 0.05 or acc $< \pi$
15: **end for**
16: **if** $\max\Delta - \min\Delta < 2$ points **then**
17:     **return** VOID {nothing moved; not a refutation}
18: **end if**
19: $c \leftarrow \big(\max\{r_\tau : \Delta_\tau < 0\},\ \min\{r_\tau : \Delta_\tau > 0\}\big)$
20: **if** c is empty **or** c is far from 0.54 **then**
21:     inspect the sweep before believing it
22: **end if**
23: $r_0 \leftarrow$ selection rate of the deployed model
24: **if** $r_0 < \min c$ **then**
25:     **return** WITHDRAWAL; add a selection-rate floor
26: **else**
27:     **return** EXTENSION; the floor buys little
28: **end if**

$r = +0.979$ and $r = +0.968$, both previously reported, are void. The same exclusion *rescued* mortgage lending, which had failed outright.

A plausible result that was arithmetic. A post-processing comparison appeared to show a clean reversal. It was void: the method re-derives its own thresholds and returned an identical model at all six operating points, so the measured effect was a constant divided by a shrinking baseline. What exposed it was an exchange rate of 0.000.

X. Limitations

Magnitude transfers within a population and not between. The crossover is nearly stable and its residual is unexplained. Coverage is four instruments, two countries and two decades, with no non-Western data. The 18-of-18 confirmation in Section VI is post-hoc. The relationship is not monotone below the crossover. The remedy is a variant, not a discovery.

XI. Discussion

The finding a practitioner can use is small and specific: **look up the rate at which your model currently says yes, and you will know whether a parity constraint is about to give or to take.**

What makes that worth saying is not that it is surprising once stated. It is that the certifying metric — the number a deployment team reports, an auditor checks, and a regulator may one day require — is identical in both worlds. A constraint that withdraws a fifth of all favourable decisions and one that creates more than it destroys produce the same certificate.

The literature already knew the metric was silent. What was missing was a way to know, in advance and from something a team already has, which of the two silences you are standing in.

Reproducibility

Every load-bearing figure is re-derived from stored results by an automated check that fails if a document and its data disagree; 28 such checks accompany this work. Pre-registrations are committed before the runs they govern.

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